-->

Voice Therapy

Number: 0646

Table Of Contents

Policy
Applicable CPT / HCPCS / ICD-10 Codes
Background
References

Policy

Scope of Policy

This Clinical Policy Bulletin addresses voice therapy.

I. **Medical Necessity**

Aetna considers voice therapy medically necessary to restore the ability of the member to produce speech sounds from the larynx for *any* of the following indications:

A. Essential voice tremor; *or*
B. Following surgery or traumatic injury to the vocal cords; *or*
C. Following treatment for laryngeal (glottic) carcinoma; *or*
D. Muscle tension dysphonia (functional dysphonia); *or*
E. Paradoxical vocal cord motion; *or*
F. Post-laryngectomy (esophageal speech production); *or*
G. Spastic (spasmodic) dysphonia; *or*
H. Symptomatic benign vocal fold lesions (cysts, nodules and polyps); *or*
I. Vocal cord paralysis.

Aetna considers voice therapy not medically necessary for *any* of the following indications:

A. Improvement of voice quality; *or*
B. Occupational or recreational purposes (e.g., public speaking, singing, etc.); *or*
C. Self-limited conditions, such as acute laryngitis.

Maintenance treatment, where the member's symptoms are not improving, is considered not medically necessary. If no clinical benefit is appreciated after 4 weeks of voice therapy, then the treatment plan should be re-evaluated. Further voice therapy is not considered medically necessary if the member does not demonstrate meaningful improvement in symptoms.

Note: Megaphones or amplifiers (e.g., ChatterVox, Mega Mite Megaphone) may be of use in the absence of illness or injury and therefore do not meet Aetna's definition of covered durable medical equipment.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Note: An electronic artificial larynx (artificial voice box) that is used by laryngectomized individuals and persons with a permanently inoperative larynx is covered as a prosthetic. See "Note" regarding electronic speech aids accompanying CPB 0437 - Speech Generating Devices. See also CPB 0560 - Voice Prosthesis for Voice Rehabilitation Following Total Laryngectomy.

Note: Voice therapy for male-to-female transgender individuals to feminize the voice or for female-to-male transgender individuals to masculinize the voice is considered not medically necessary and cosmetic. See also CPB 0615 - Gender Affirming Surgery.

II. Experimental, Investigational, or Unproven

Aetna considers voice therapy experimental, investigational, or unproven for the following because their effectiveness has not been established:

- Pre-operative voice therapy for improving outcomes of surgery for benign vocal fold lesions;
- Vocal fold motion impairment following chemotherapy administration.

III. Policy Limitations and Exclusions

Note: Voice therapy is subject to any benefit limitations and exclusions applicable to speech therapy.

IV. Related Policies

- CPB 0243 - Speech Therapy
- CPB 0437 - Speech Generating Devices
- CPB 0560 - Voice Prosthesis for Voice Rehabilitation Following Total Laryngectomy
- CPB 0615 - Gender Affirming Surgery

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
92507	Treatment of speech, language, voice, communication, and/or auditory processing disorder; individual [voice therapy] [not covered for male-to-female transgender individuals to feminize the voice or female-to-male transgender individuals to masculinize the voice]
92508	Treatment of speech, language, voice, communication, and/or auditory processing disorder; group, two or more individuals [voice therapy] [not covered for male-to-female transgender individuals to feminize the voice or female-to-male transgender individuals to masculinize the voice]
Other CPT codes related to the CPB:	
31360	Laryngectomy; total, without radical neck dissection
31365	total, with radical neck dissection

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
31367	subtotal supraglottic, without radical neck dissection
31368	subtotal supraglottic, with radical neck dissection
31370	Partial laryngectomy (hemilaryngectomy); horizontal
31375	laterovertical
31380	anterovertical
31382	antero-latero-vertical
31505 - 31592	Laryngoscopy and laryngoplasty procedures (therapeutic)
31611	Construction of tracheoesophageal fistula and subsequent insertion of an alaryngeal speech prosthesis (e.g., voice button, Blom-Singer prosthesis)
92521	Evaluation of speech fluency (eg, stuttering, cluttering)
92522	Evaluation of speech sound production (eg, articulation, phonological process, apraxia, dysarthria) [not covered for male-to-female transgender individuals to feminize the voice or female-to-male transgender individuals to masculinize the voice]
92523	Evaluation of speech sound production (eg, articulation, phonological process, apraxia, dysarthria); with evaluation of language comprehension and expression (eg, receptive and expressive language) [not covered for male-to-female transgender individuals to feminize the voice or female-to-male transgender individuals to masculinize the voice]
92524	Behavioral and qualitative analysis of voice and resonance [not covered for male-to-female transgender individuals to feminize the voice or female-to-male transgender individuals to masculinize the voice]
92597	Evaluation for use and/or fitting of voice prosthetic device to supplement oral speech
96401 - 96417	Injection and intravenous infusion chemotherapy and other highly complex drug or highly complex biologic agent administration
96420 - 96425	Intra-arterial chemotherapy and other highly complex drug or highly complex biologic agent administration

HCPCS codes covered if selection criteria are met:

G0153	Services performed by a qualified speech-language pathologist in the home health or hospice setting, each 15 minutes
G0161	Services performed by a qualified speech-language pathologist, in the home health setting, in the establishment or delivery of a safe and effective speech-language pathology maintenance program, each 15 minutes
S9128	Speech therapy, in the home, per diem

HCPCS codes not covered for indications listed in the CPB:

L8510	Voice amplifier

Other HCPCS codes related to the CPB:

L8500	Artificial larynx, any type
L8505	Artificial larynx replacement battery / accessory, any type
L8507	Tracheo-esophageal voice prosthesis, patient inserted, any type, each
L8509	Tracheo-esophageal voice prosthesis, inserted by a licensed health care provider, any type
V5362	Speech screening

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
V5363	Language screening

ICD-10 codes covered if selection criteria are met:

D14.1	Benign neoplasm of larynx [benign vocal fold lesions]
F44.4	Conversion disorder with motor symptom or deficit [functional dysphonia]
G25.2	Other specified forms of tremor [voice]
J38.00 - J38.02	Paralysis of vocal cords and larnyx
J38.1	Polyp of vocal cord and larynx
J38.2	Nodules of vocal cords
R49.0	Dysphonia [functional dysphonia]
S19.83X+	Other specified injuries of vocal cord
Z90.02	Acquired absence of larynx [Esophageal speech production]

ICD-10 codes not covered for indications listed in the CPB:

A15.5	Tuberculous laryngitis
A36.2	Laryngeal diphtheria
A52.73	Symptomatic late syphilis of other respiratory organs
A69.1	Other Vincent's infections [Vincent's angina]
F64.1	Gender identity disorder in adolescences and adulthood
I69.020 - I69.028 I69.120 - I69.128 I69.220 - I69.228 I69.320 - I69.328 I69.820 - I69.828 I69.920 - I69.928	Sequelae of cerebrovascular disease, speech and language deficits
J02.0	Streptococcal pharyngitis
J03.00	Acute streptococcal tonsillitis
J03.01	Acute recurrent streptococcal tonsillitis
J04.0	Acute laryngitis
J04.2	Acute laryngotracheitis
J05.0	Acute obstructive laryngitis [croup]
J06.0	Acute laryngopharyngitis
J10.1	Influenza due to other identified influenza virus with other respiratory manifestations
J11.1	Influenza due to unidentified influenza virus with other respiratory manifestations
R47.02	Dysphasia
R47.1	Dysarthria and anarthria
R47.81 - R47.89	Other speech disturbances
Z51.89	Encounter for other specified aftercare [occupational therapy]

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Background

Vocal cord paralysis is a voice disorder that occurs when one or both of the vocal cords (or vocal folds) do not open or close properly (NIDCD, 1999). Vocal cord paralysis is a common disorder, and symptoms can range from mild to life-threatening. Someone who has vocal cord paralysis often has difficulty swallowing and coughing because food or liquids slip into the trachea and lungs. This happens because the paralyzed cord or cords remain open, leaving the airway passage and the lungs unprotected.

Vocal cord paralysis may be caused by head trauma, a neurological insult such as a stroke, a neck injury, lung or thyroid cancer, a tumor pressing on a nerve, or a viral infection (NIDCD, 1999). In older people, vocal cord paralysis is a common problem affecting voice production. People with certain neurological conditions, such as multiple sclerosis or Parkinson's disease, or people who have had a stroke may experience vocal cord paralysis. In many cases, however, the cause is unknown.

People who have vocal cord paralysis experience abnormal voice changes, changes in voice quality, and discomfort from vocal straining (NIDCD, 1999). For example, if only 1 vocal cord is damaged, the voice is usually hoarse or breathy. Changes in voice quality, such as loss of volume or pitch, may also be noticeable. Damage to both vocal cords, although rare, usually causes people to have difficulty breathing because the air passage to the trachea is blocked.

Vocal cord paralysis is usually diagnosed by an otolaryngologist (NIDCD, 1999). Noting the symptoms the patient has experienced, the otolaryngologist will ask how and when the voice problems started in order to help determine their cause. Next, the otolaryngologist listens carefully to the patient's voice to identify breathiness or harshness. Then, using an endoscope, the otolaryngologist looks directly into the throat at the vocal cords. A speech-language pathologist may also use an acoustic spectrograph, an instrument that measures voice frequency and clarity, to study the patient's voice and document its strengths and weaknesses.

There are several methods for treating vocal cord paralysis, among them surgery and voice therapy. In some cases, the voice returns without treatment during the first year after damage (NIDCD, 1999). For that reason, doctors often delay corrective surgery for at least 1 year to be sure the voice does not recover spontaneously. During this time, the suggested treatment is usually voice therapy, which may involve exercises to strengthen the vocal cords or improve breath control during speech. Sometimes, a speech-language pathologist must teach patients to talk in different ways. For instance, the therapist might suggest that the patient speak more slowly or consciously open the mouth wider when speaking.

Surgery involves adding bulk to the paralyzed vocal cord or changing its position (NIDCD, 1999). To add bulk, an otolaryngologist injects a substance, commonly Teflon, into the paralyzed cord. Other substances currently used are collagen, silicone, and body fat. The added bulk reduces the space between the vocal cords so the non-paralyzed cord can make closer contact with the paralyzed cord and thus improve the voice.

Sometimes an operation that permanently shifts a paralyzed cord closer to the center of the airway may improve the voice (NIDCD, 1999). Again, this operation allows the non-paralyzed cord to make better contact with the paralyzed cord. Adding bulk to the vocal cord or shifting its position can improve both voice and swallowing. After these operations, patients may also undergo voice therapy, which often helps to fine-tune the voice.

Treating people who have 2 paralyzed vocal cords may involve performing a surgical procedure called a tracheotomy to help breathing (NIDCD, 1999). In a tracheotomy, an incision is made in the front of the patient's neck and a breathing tube (tracheotomy tube) is inserted through a hole, called a stoma, into the trachea. Rather than breathing through the nose and mouth, the patient now breathes through the tube. Following surgery, the patient may need therapy with a speech-language pathologist to learn how to care for the breathing tube properly and how to reuse the voice.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Vocal cord nodules (singer's nodules) are small, hard, callus like growths that usually appear singly on the vocal cord (Merck, 1997). Nodules consist of condensations of hyaline connective tissue in the lamina propria at the junction of the anterior 1/3 and posterior 2/3 of the free edges of the true vocal cords. Vocal cord polyps are small, soft growths that usually appear singly on a vocal cord. They are most often caused by vocal abuse or long-term exposure to irritants, such as chemical fumes or cigarette smoke.

Vocal cord nodules are caused by chronic voice abuse, such as yelling, shouting, or using an unnaturally low frequency (Merck, 1997). Chronic infections caused by allergies and inhalation of irritants, such as cigarette smoke, may also produce these lesions. Hoarseness and a breathy voice result. Carcinoma should be excluded by biopsy.

Treatment for nodules that do not resolve with voice therapy involves surgical removal of the nodules at direct laryngoscopy and correction of the underlying voice abuse. Vocal nodules in children usually regress with voice therapy alone.

If nothing is done to change vocal abuse habits, vocal cord nodules can last a lifetime, and may even recur after surgical removal. With proper voice training with a certified therapist, nodules can disappear with 6 to 10 voice therapy sessions over 6 to 12 weeks. With rest, some vocal cord polyps will go away on their own within a few weeks. Most, however, will require surgical removal.

Speech therapists use a variety of techniques to restore a patient's ability to produce speech, including:

- Patients are instructed in voice modification and relative voice rest. At its most extreme, relative voice rest involves an initial period of between 4 and 7 days using the voice no more than 15 mins in each 24-hr period. It is normally undertaken with speech therapist supervision, once its advantages and disadvantages have been explored with the patient.
- Patients are instructed to minimizing voice use. This involves speaking no more than is absolutely necessary. Patients are taught to be succinct when speaking, to avoid any loud use of the voice, to keep phone calls brief and to avoid all non-speech voice use (throat clearing, coughing, "voiced" sneezing, crying, "voiced" laughing, and odd sound-effects).
- Patients are told to avoid any "voice abuse" practices identified by the speech therapist. Patients are instructed to avoid or modify internal (bodily and psychological) and external (environmental) voice-damaging environments as much as possible.
- Patients are instructed in how to engage in a short warm-up period of controlled, soft vocal exercises before using the voice. Patients may also be advised, where appropriate, to keep a supply of drinking water handy, and to massage under their chin if their mouth becomes dry.
- Patients are taught good voice production techniques. Patients may be instructed in optimal breathing patterns, to speak more slowly, to articulate clearly, to speak at a comfortable pitch and loudness level, to use pitch change rather than volume change for emphasis, to monitor their posture, to avoid monotone delivery, and to be aware of muscle tension.
- Patients may be advised to avoid speaking when they are ill or tired. Dehydration, fatigue and other general medical conditions have an effect on the mucosa covering the vocal cords, potentially altering lubrication and vocal efficiency.

Laryngitis due to viral infection usually resolves within 1 to 3 weeks. Laryngitis due to vocal abuse will generally go away on its own in a few days with voice rest.

Functional voice disorders are characterized by the presence of vocal symptoms without anatomical laryngeal abnormality. Muscle tension dysphonia (MTD) is the most common disorder in this category.

Van Houtte et al (2013) stated that muscle tension dysphonia (MTD) is a clinical and diagnostic term describing a spectrum of disturbed vocal fold behavior caused by increased tension of the (para)laryngeal musculature. Recent knowledge introduced MTD as a bridge between functional and organic disorders. These researchers addressed the causal and contributing factors of MTD and evaluated the different treatment options. They searched Medline (PubMed, 1950 to 2009) and CENTRAL (The Cochrane Library, Issue 2, 2009). Studies were included if they reviewed the classification of functional dysphonia or the

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

pathophysiology of MTD. Etiology and pathophysiology of MTD and circum-laryngeal manual therapy (CMT) were obligatory based on reviews and prospective cohort studies because randomized controlled trials (RCTs) are non-existing. Concerning the treatment options of voice therapy and vocal hygiene, selection was based on RCTs and systematic reviews. Etiological factors can be categorized into 3 new subgroups:

I. psychological and/or personality factors,
II. vocal misuse and abuse, and
III. compensation for underlying disease.

I. psychological and/or personality factors,
II. vocal misuse and abuse, and
III. compensation for underlying disease.

The effective treatment options for MTD are

I. indirect therapy: vocal hygiene and patient education;
II. direct therapy: voice therapy and CMT;
III. medical treatment; and
IV. surgery for secondary organic lesions.

I. indirect therapy: vocal hygiene and patient education;
II. direct therapy: voice therapy and CMT;
III. medical treatment; and
IV. surgery for secondary organic lesions.

The authors concluded that MTD is the pathological condition in which an excessive tension of the (para)laryngeal musculature, caused by a diverse number of etiological factors, leads to a disturbed voice. Etiological factors range from psychological/personality disorders and vocal misuse/abuse to compensatory vocal habits in case of laryngopharyngeal reflux, upper airway infections, and organic lesions. MTD needs to be approached in a multi-disciplinary setting where close co-operation between a laryngologist and a speech language pathologist is possible.

Roy et al (2005) reported on the prevalence of voice disorders from a questionnaire of 1,326 adults in Iowa and Utah. The authors reported a lifetime prevalence of a voice disorder of 29.9 %, with 6.6 % of participants reporting a current voice disorder. Risk factors for chronic voice disorders included sex (women), age (40 to 59 years), voice use patterns and demands, esophageal reflux, chemical exposures, and frequent cold/sinus infections. However, tobacco or alcohol use did not independently increase the odds of reporting of a chronic voice disorder. Altman et al (2005) reported on a retrospective chart review of 150 patients with MTD seen at a single referral center over 30-month period. The authors stated that significant factors in patient history believed to contribute to abnormal voice production were gastroesophageal reflux in 49 %, high stress levels in 18%, excessive amounts of voice use in 63%, and excessive loudness demands on voice use in 23 %. Otolaryngologic evaluation was performed in 82 % of patients, in whom lesions, significant vocal fold edema, or paralysis/paresis was identified in 52.3 %. Speech pathology assessment revealed poor breath support, inappropriately low pitch, and visible cervical neck tension in the majority of patients. Inappropriate intensity was observed in 23.3 % of patients.

Morrison et al (1983) stated that 100 out of 500 consecutive patients seen at the Voice Clinic of the University of British Columbia exhibited features of muscle tension dysphonia. The authors stated that muscular tension dysphonia is commonly seen in young and middle aged females, and is manifest by excess tension in the para-laryngeal and suprahyoid muscles, an open posterior glottic chink, larynx rise, and frequently mucosal changes on the vocal cords.

Much of the literature on muscle tension dysphonia has focused on examining differences between affected persons and normal controls. Zheng et al (2012) reported on differences in a variety of aerodynamic parameters in persons with MTD and normal controls. Lowell et al (2012) reported on x-ray measures of the hyoid and larynx in persons diagnosed with MTD and normal controls. There is inconsistent evidence for

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

differences in surface EMG measures between persons affected with MTD and normal controls (van Houtte et al, 2011; Hocevar-Boltezar et al, 1998). In a prospective control-blinded, cross-sectional study of 51 patients with functional dysphonia and 52 non-dysphonic controls, Sama et al (2001) found that the laryngoscopy features commonly associated with functional dysphonia are frequently prevalent in the non-dysphonic population and fail to distinguish patients with functional dysphonia from normal subjects.

Nguyen et al (2009) reported on differences in presenting symptoms of MTD in Vietnamese-speaking teachers compared to the typical symptoms in English-speaking persons, suggesting a potential contribution of linguistic-specific factors and teacher-specific factors to the presentation of MTD. Rubin et al (2007) reported that an ear-nose and throat surgeon identified certain patterns of musculoskeletal abnormalities in 26 voice professionals with voice disorders, including a high held larynx, a shortening or contraction of the stylohyoid and sternocleidomastoid muscles, and a weak deep flexor mechanism. The authors noted that the correlation proved to be excellent among the subgroup of patients who were performers, but only fair-to-good among the other voice professionals. The investigators stated that, in this small group, most patients seemed to improve with physiotherapy, although it was not clear that these improvements could be attributed to physiotherapy, as the patients concurrently received other treatments.

Kooijman et al (2005) reported on observed relationships between extrinsic laryngeal muscular hypertonicity and deviant body posture on the one hand and voice handicap and voice quality on the other hand in 25 teachers with persistent voice complaints and a history of voice-related absenteeism. The authors states that muscular tension and body posture should be assessed in persons presenting with voice disorders.

Other literature has focused on differences between persons with MTD and adductor spasmodic dysphonia (ADSD). Patel et al (2011) reported on differences in vibratory characteristics between persons with MTD and persons with ADSD. In a case-control study, Houtz et al (2010) reported spectral noise differences between females with MTD and ADSD, but not for men. Roy et al (2008) reported on differences in phonatory breaks between persons with ADSD and persons with MTD. Angsuwarangsee and Morrison (2002) reported on findings on palpation of extrinsic laryngeal muscles in muscle misuse dysphonia (MMD) with or without gastroesophageal reflux. Sapienza et al (2000) reported on differences in acoustic phonatory events in patients with ADSD and MTD. Higgins et al (1999) reported significant differences in mean phonatory air flow in subjects with ADSD, MTD, and normal subjects, and very large inter-subject variation in mean phonatory air flow for both subjects with ADSD and MTD. Roy (2010) stated that differences between MTD and ADSD have been identified during fiberoptic laryngoscopy, phonatory airflow measurement, acoustic analysis, and variable sign expression based upon phonatory task. He reported, however, that "no single diagnostic test currently exists that reliably distinguishes the two disorders."

Some of the literature on voice therapy focuses on the use of laryngeal manipulation in the diagnosis and treatment of voice disorders (e.g., Roy et al, 1996; Rubin et al, 2000). Roy et al (1996) stated that excessive activity of the extra-laryngeal muscles affects laryngeal function and contributes to a spectrum of interrelated symptoms and syndromes including muscle tension dysphonia and spasmodic dysphonia. The authors explained that manual laryngeal musculoskeletal tension reduction techniques are used in the diagnosis and management of laryngeal hyperfunction syndromes. The authors explained that manual technique consists of focal palpation to determine

I. extent of laryngeal elevation,
II. focal tenderness,
III. voice effect of applying downward pressure over the superior border of the thyroid lamina, and
IV. extent of sustained voice improvement following circum-laryngeal massage.

I. extent of laryngeal elevation,
II. focal tenderness,
III. voice effect of applying downward pressure over the superior border of the thyroid lamina, and
IV. extent of sustained voice improvement following circum-laryngeal massage.

Roy (2003) stated that while voice therapy by an experienced speech-language pathologist remains an effective short-term treatment for functional dysphonia (FD) in the majority of cases, but less is known

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

regarding the long-term fate of such intervention. The author stated that poorly regulated activity of the intrinsic and extrinsic laryngeal muscles is cited as the proximal cause of functional dysphonia, but the origin of this dyregulated laryngeal muscle activity has not been fully elucidated. The author stated that several causes have been cited as contributing to this imbalanced muscle tension. Roy stated, however, that recent research evidence points to specific personality traits as important contributors to its development and maintenance. Roy stated that further research is needed to better understand the pathogenesis of functional dysphonia, and factors contributing to its successful management.

Many of the studies of report on patients before and after therapy, but lack a control group (e.g., Stepp et al, 2011; Lee and Son, 2005; Birkent et al, 2004). Other evidence voice therapy in muscle tension dysphonia consists of case reports (Roy et al, 1996).

Stepp et al (2011) found that relative fundamental frequency (RFF) surrounding a voiceless consonant in patients with hyper-functionally related voice disorders increased towards patterns of healthy normal individuals following a course of voice therapy. Pre- and post-therapy measurements of RFF were compared in 16 subjects undergoing voice therapy for voice disorders associated with vocal hyperfunction. The authors reported that a 2-way analysis of variance showed a statistically significant effect of both cycle of vibration near the consonant and therapy phase (pre- versus post-). A post-hoc paired Student's t-test showed that posttherapy RFF measurements were significantly higher (more typical) than pretherapy measurements. Limitations of this study include its small size, lack of control group, limited followup, and use of intermediate endpoints.

Lee and Son (2005) reported on a retrospective review of the clinical records of 8 Korean male children diagnosed as having MTD, 7 of whom had bilateral vocal nodules on laryngoscopic examination. The authors reported that a few sessions of voice therapy, focusing on awareness, relaxation, respiration and easy-onset phonation to reduce the tension around the laryngeal muscles, resulted in dramatic improvement of their voice quality and pitch adjustment. Limitations of this study included its small size, retrospective nature, and lack of control group.

Rosen et al (2000) reported on a retrospective review the results of treatment of a series of 37 subjects with unilateral vocal fold paralysis (UVFP) (n = 14), muscle tension dysphonia (MTD) (n = 10), and vocal fold polyp or cyst (VFP/C) (n = 13). The majority of the UVFP group were treated with surgical vocal fold medialization followed by postoperative voice therapy. Patients with MTD were treated with voice therapy. Patients in the VFP/C group were treated with phono-microsurgery and pre- and post-surgical voice therapy. All patients in the study were also treated with combined behavioral and medical therapy for suspected laryngopharyngeal reflux when appropriate. For patients with MTD, voice therapy consisted of 6 to 14 sessions, with an average of 8 sessions. The primary endpoint of the study was the Voice Handicap Inventory (VHI), which quantifies the patients' perception of disability due to voice difficulties. The authors found a statistically significant difference between the pre-and post-treatment VHI for the UVFP and the VFP/C groups. There was a trend toward improvement in VHI in the MTD group, but it lacked statistical significance.

Birkent et al (2004) reported on the efficacy of voice therapy techniques in 37 Turkish patients with functional dysphonia, including mutational falsetto (n = 16), vocal nodule (n = 17), and muscle-tension dysphonia (n = 4). The authors reported that 15 patients with mutational falsetto (93 %), 8 patients with vocal nodules (47 %), and all 4 patients with muscle-tension dysphonia were cured by voice therapy techniques. Limitations of this study include its small size and lack of control group.

Rubin et al (2007) found that persons presenting with voice problems frequently have musculoskeletal issues. All 26 patients with presenting voice problems were found by a physiotherapist to have musculoskeletal abnormalities. Certain patterns of musculoskeletal abnormalities were frequently encountered, including a high held larynx, a shortening or contraction of the stylohyoid and sternocleidomastoid muscles, and a weak deep flexor mechanism. The authors stated that the correlation proved to be excellent among the subgroup of performers, but only fair-to-good among the other voice professionals. The authors stated that, in this small group, most patients seemed to improve following

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

physiotherapy, although the authors stated that "it must be noted that management was not limited to physiotherapy."

Andrews et al (1986) evaluated two methods of relaxing laryngeal musculature in 10 adults with hyper-functional dysphonia. Subjects were matched into groups receiving either laryngeal surface EMG biofeedback or progressive relaxation, both within a graded voice training program. Assessments were conducted pre-treatment, post-treatment and at 3- month follow-up. Measures included the level of superficial laryngeal tension using an electromyogram, control of vocal fold vibration from an electrolaryngograph and an auditory evaluation using a phonation profile. Two personality questionnaires were administered and the subjects'self-rating of voice was recorded. The authors reported a significant improvement in all measures for both programs, which was maintained at 3-month follow-up, with no significant differences between the two approaches. Limitations of the study include small sample size, lack of control group, and short duration of followup.

Fex et al (1994) reported on the effectiveness of voice therapy using the accent method in 10 patients who were referred to a speech pathologist for functional voice disorders. Three of the 10 patients had bilateral nodules. The voices were analyzed acoustically before and after treatment. Of the parameters tested, pitch perturbation quotient, amplitude perturbation quotient, normalized noise energy for 1 to 4 kHz, and fundamental frequency showed significant improvement. Limitations of this study include the small sample size, lack of control group, and lack of information on the durability of treatment effects.

Kitzing and Akerlund (1993) found only weakly positive correlations between perceptual improvements in voice quality and long-time average voice spectrograms with voice therapy in 174 subjects with non-organic voice disorders (functional dysphonia).

Carling et al (1999) reported on a study where 45 patients diagnosed as having nonorganic dysphonia were assigned in rotation to 1 of 3 groups. Patients in group 1 received no treatment and acted as a control group. Patients in groups 2 and 3 received a program of indirect therapy and direct with indirect therapy, respectively. A range of qualitative and quantitative measures were carried out on all patients before and after treatment to evaluate change in voice quality over time. All patients were assessed before their treatment program began. Following the treatment period the patients were assessed a second time and then again at a review 1 month later. The difference between the first and third assessments was used to measure the degree of change. The authors reported that there was a significant difference between the 3 treatment groups in the amount of change for the voice severity, electro-laryngograph, and shimmer measurements and on ratings provided by a patient questionnaire ($p < 0.05$). However, other measures failed to show significant differences between the 3 groups. Limitations of this study include a suboptimal method of group assignment, failure to control the effect of contact time on outcomes, lack of data on durability of results, and the use of only one single therapist and patient selection criteria that may influence the generalizability of the results. In addition, the study was not designed to assess which aspects of the therapy that the subjects received was most effective.

Systematic evidence reviews have cited the need for additional research into the effectiveness of voice therapy for MTD. In a systematic evidence review of voice therapy, Speyer (2008) reported that, due to the small number of published treatment outcome studies and the methodological heterogeneity among published studies, very few conclusions relative to the effectiveness of voice therapy ‘‘in general’’ may be drawn from the literature. From the results of his review, Speyer suggests a tendency for positive intervention outcomes to be more commonly reported for

I. very specific therapy approaches such as manual laryngeal tension reduction or the Accent method of voice therapy, and
II. for studies focusing on specific clinical populations such as mutational falsetto or vocal nodules.

I. very specific therapy approaches such as manual laryngeal tension reduction or the Accent method of voice therapy, and
II. for studies focusing on specific clinical populations such as mutational falsetto or vocal nodules.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Mathesion (2011) reported that studies of laryngeal manual therapies for MTD have reported positive effects, "but the evidence base remains extremely small." The author concluded that "a higher level of evidence is required, including randomized controlled trials, to investigate its role in comparison with other interventions." Bos-Clark and Carding (2011) reviewed the recent literature since a 2009 Cochrane review regarding the effectiveness of voice therapy for patients with functional dysphonia. The authors found a range of articles report on the effects of voice therapy treatment for functional dysphonia, with a wide range of interventions described. The authors only one randomized controlled trials. The authors noted that, "in primary research, methodological issues persist: studies are small, and not adequately controlled." The authors noted that these more recent studies show improved standards of outcome measurement and of description of the content of voice therapy. The authors concluded: "There is a continued need for larger, methodologically sound clinical effectiveness studies. Future studies need to be replicable and generalizable in order to inform and elucidate clinical practice."

Van Lierde et al (2004) reported on a "pilot study" of manual circum-laryngeal therapy (MCT) in 4 Dutch professional voice users with a persistent MTD. Subjects were evaluated before and 1 week after completion of therapy. The authors reported that all 4 subjects showed improvement in perceptual vocal quality and dysphonia severity index (DSI) after 25 sessions of MTD. Limitations of the study include its small size, lack of control group, and lack of evidence of durability.

Van Lierde et al (2010) measured the dysphonia severity index in 10 subjects before and after treatment with 45 minutes of vocalization with abdominal breath support, followed by 45 minutes of manual circum-laryngeal therapy (MCT). The authors found no significant improvements in the dysphonia severity index before and after vocalization with abdominal breath support, and significant differences before and after MCT. Limitations of this study include its small size, pre-post design, lack of measurement of clinical outcomes, and lack of evidence on durability of treatment results.

Mathieson et al (2009) reported on the results of a "pilot study" before and after 1 week of MCT in 10 subjects with MTD. The authors reported that there was a significant reduction in average pertubation during connected speech by acoustical analysis, and a reduction in the severity and frequency of VTD (a new perceptual, self-rating scale) after treatment. Limitations of this study include its small size, lack of control group, and lack of evidence of durability of results.

In a retrospective study that the authors considered "preliminary", Roy et al (2009) documented significant changes in vowel acoustic measures after MCT in 111 women with MTD, suggesting improvement in speech articulation with MCT. Limitations of the study include its pre-post design, lack of control group, and lack of data on the durability of treatment. The same investigator group found differences in changes in both articulatory and phonatory behavior in these 111 women following MCT (Dromey et al, 2008).

Roy et al (1997) reported on the use of MCT in 25 consecutive female patients with functional dysphonia. Pre- and post-treatment audio recordings of connected speech and sustained vowel samples were submitted to auditory-perceptual and acoustical analysis to assess the effects of a single treatment session. To complement audio recordings, subjects were interviewed in follow-up regarding the stability of treatment effects. One post-treatment measurement occurred at a mean duration of 3.6 months following initial treatment, and a second post-treatment measurement was obtained at a mean duration of 16.5 months. Pre- and post-treatment comparisons demonstrated significant voice improvements. Pre-treatment severity ratings for connected-speech samples were reduced from a mean of 5.37 to 1.91 immediately following the management session ($p = 0.0001$). Comparison of mean pre-treatment severity scores with each subsequent post-treatment follow-up mean revealed significant differences ($p < 0.0004$). No significant differences were identified when comparing among post-treatment means (post 1- versus post 2-; post 1- versus post 3-; and post 2- versus post 3-). Interviews revealed 68 % of subjects reported occasional self-limiting partial recurrences, with 3/4 of relapses occuring within 2 months of completing therapy. The study was limited by small sample size and lack of control group. In addition, 9 of 25 subjects were lost to long-term followup. The authors noted: "The finding that patients responded to a single treatment session is encouraging, but a serious question lingers as to whether improvements in voice are a consequence of reduced laryngeal musculoskeletal tension or are due to factor(s) unrelated to the manual technique". The authors stated: "The absence of a nontreatment or alterative treatment control group leaves open the possibility of numerous

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

alterative explanations for the observed treatment effects, including placebo effects, the clinician's instructions, expectations, experience, and confidence".

Roy and Hendarto (2005) found no significant changes in mean speaking fundamental frequency (SFF) after MCT in 40 women with functional dysphonia, despite subjective reports of improvement after therapy.

Roy and Leeper (1993) reported on the results of MCT in 17 patients with functional dysphonia. The effects of the therapy regimen were analyzed using perceptual and acoustical measures of vocal function. The results of an assessment immediately after treatment indicated a significant change in the direction of "normal" vocal function in the majority of patients within one treatment session. Perceptual measures of severity were consistently more likely to be rated as normal following treatment. Acoustic measures of voice confirmed significant improvements in jitter, shimmer, and signal-to-noise ratio (SNR). Following a telephone interview procedure one week posttreatment, it was subjectively determined that of the 14 patients who were rated as demonstrating either normal voice or only the mildest dysphonic symptoms immediately following treatment, 13 patients (93%) were judged by the principal investigator to have maintained the improved vocal quality. Limitations of the study include the small sample size, lack of control group, and lack of data on durability of treatment effects.

Van Houtte et al (2011) found that results of studies of CMT for MTD are "promising"; the studies were small and no randomized controlled clinical trials were conducted. "Further research is necessary to compare this treatment to voice therapy to delineate the patient group that would benefit from this therapy, establish the number of session of CMT that are necessary to maintain the improved voice quality".

In regards to a multi-dimensional protocol for assessing functional results of voice therapy, Dejonckere (2000) noted that results show that there is a large variation in the inter-individual and inter-dimensional results of the voice therapy (4 to 26 sessions) – in the same patient, one dimension may be significantly improved while another one is significantly worsened.

Voice therapy has been shown to be effective in rehabilitating persons treated for early glottic carcinoma. In a randomized, controlled trial, van Gogh and colleagues (2006) evaluated the effectiveness of voice therapy in patients who experienced voice problems after receiving treatments for early glottic cancer. Of 177 patients, 6 to 120 months after treatment for early glottic carcinoma, 70 patients (40 %) suffered from voice impairment based on a 5-item screening questionnaire. About 60 % of those 70 patients were not interested in participating in the study. A total of 23 patients who were willing to participate were randomly assigned either to a voice therapy group (n = 12) or to a control group (n = 11). Multi-dimensional voice analyses (the self-reported Voice Handicap Index [VHI], acoustic and perceptual voice quality analysis, videolaryngostroboscopy, and the Voice Range Profile) were conducted twice: before and after voice therapy or with 3 months in between for the control group. Statistical analyses of the difference in scores (post-measurement minus pre-measurement) showed significant voice improvement after voice therapy on the total VHI score, percent jitter, and noise-to-harmonics ratio in the voice signal and on the perceptual rating of vocal fry. The authors concluded that voice therapy proved to be effective in patients who had voice problems after treatment for early glottic carcinoma. Improvement not only was noticed by the patients (VHI) but also was confirmed by objective voice parameters.

Paradoxical vocal fold motion is characterized as an abnormal adduction of the vocal cords during the respiratory cycle (especially during the inspiratory phase) that produces airflow obstruction at the level of the larynx (Buddiga, 2010). Paradoxical vocal cord motion frequently mimics persistent asthma. The localization of airflow obstruction to the laryngeal area is an important clinical discriminatory feature in patients with paradoxical vocal cord motion. Pulmonary function testing is the most useful tool in discriminating between paradoxical vocal cord motion and asthma. Flow-volume loops typically demonstrate inspiratory loop flattening, i.e., an inspiratory flow decrease during symptomatic periods suggestive of paradoxical vocal cord motion. The hallmark of diagnosis is noted on direct rhino-laryngoscopy; a glottic chink is present along the posterior portion of the vocal cords, while the anterior portion of the vocal cords is adducted (Buddiga, 2010). The mainstay of treatment for paradoxical vocal cord motion involves teaching the patient vocal cord relaxation techniques and breathing exercises.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Young and Blitzer (2007) noted that spasmodic dysphonia is a disabling disorder of the voice characterized primarily by involuntary disruptions of phonation. Botulinum toxin injections of the thyroarytenoid muscles have been the treatment of choice for adductor spasmodic dysphonia (ADSD). In a case-series study, these researchers described a new technique to address the problem of compensatory or supraglottic hyper-adduction in some of these patients. A total of 4 patients with ADSD with sphincteric supraglottic contraction were seen for evaluation of botulinum toxin injection. On fiberoptic examination, it was noted that they had type I hyperadduction of the true vocal cords with a significant type III, and/or type IV hyperadduction of the supraglottis. After standard management of the thyroarytenoid muscles, the strained/strangled voice continued. On fiberoptic examination it was noted that the vocal folds were weakened, but the supraglottic hyperfunction persisted. Patients were treated by speech therapists to unload their supraglottis without success. All patients then had their oblique portion of the lateral cricoarytenoid muscles injected with botulinum toxin A through a thyrohyoid approach. This was done in the office under electromyographic control. On follow-up, all patients demonstrated improvement in the quality of their voices (as compared to thyroarytenoid injections alone). The authors described a new technique for injection of the supraglottic portion of the lateral cricoarytenoid muscles. They demonstrated this can be done safely and successfully in an office setting with electromyography control.

In a Cochrane review, Ruotsalainen et al (2007) evaluated the effectiveness of interventions to treat functional dysphonia in adults. Randomized controlled trials of interventions evaluating the effectiveness of treatments targeted at adults with functional dysphonia were included in this analysis. For work-directed interventions interrupted time series and prospective cohort studies were also eligible. Two authors independently extracted data and assessed trial quality. Meta-analysis was performed where appropriate. These investigators identified 6 RCTs including a total of 163 participants in intervention groups and 141 controls. One trial was high quality. Interventions were grouped into

I. direct voice therapy,
II. indirect voice therapy,
III. combination of direct and indirect voice therapy ,and
IV. other treatments: pharmacological treatment and vocal hygiene instructions given by phoniatrist.

I. direct voice therapy,
II. indirect voice therapy,
III. combination of direct and indirect voice therapy ,and
IV. other treatments: pharmacological treatment and vocal hygiene instructions given by phoniatrist.

No studies were found evaluating direct voice therapy on its own. One study did not show indirect voice therapy on its own to be effective when compared to no intervention. There is evidence from 3 studies for the effectiveness of a combination of direct and indirect voice therapy on self-reported vocal functioning (SMD -1.07; 95 % confidence interval [CI]: -1.94 to -0.19), on observer-rated vocal functioning (WMD -13.00; 95 % CI: -17.92 to -8.08) and on instrumental assessment of vocal functioning (WMD -1.20; 95 % CI: -2.37 to -0.03) when compared to no intervention. The results of 1 study also show that the remedial effect remains significant for at least 14 weeks on self-reported vocal functioning (SMD -0.51; 95 % CI: -0.87 to -0.14) and on observer-rated vocal functioning (Buffalo Voice Profile) (WMD -0.80; 95 % CI: -1.14 to -0.46). There is also limited evidence from 1 study that the number of symptoms may remain lower for 1 year. The combined therapy with biofeedback was not shown to be more effective than combined therapy alone in 1 study nor was pharmacological treatment found to be more effective than vocal hygiene instructions given by phoniatrist in 1 study. Publication bias may have influenced the results. The authors concluded that evidence is available for the effectiveness of comprehensive voice therapy comprising both direct and indirect therapy elements. Effects are similar in patients and in teachers and student teachers screened for voice problems. Moreover, they stated that larger and methodologically better studies are needed with outcome measures that match treatment aims. Commenting on the Cochrane review, Carding (2011) has stated that, "[i]n contrast to popular opinion, the evidence base that underpins voice therapy practice remains incomplete and inconclusive".

Eastwood et al (2015) performed a systematic review of behavioral intervention for the treatment of adults with muscle tension voice disorders (MTVD). A search of 12 electronic databases and reference lists for

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

studies published between the years 1990 to 2014 was conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Inclusion and exclusion criteria included type of publication, participant characteristics, intervention, outcome measures and report of outcomes. Methodological quality rating scales and confidence in diagnostic scale supported the literature evaluation. A total of 7 papers met the inclusion criteria. Significant improvement on at least 1 outcome measure was reported for all studies. Effect sizes were small-to-large. Methodological qualities of research were varied. No study explicitly reported treatment fidelity and cumulative intervention intensity could only be calculated for 2 out of 7 studies. Outcome measures were used inconsistently and less than 50 % of the measures had reported reliability values. Confidence in the accuracy of subject diagnosis on average was rated as low. Specific "active ingredients" for therapeutic change were not identified. The authors concluded that voice therapy for the treatment of MTVD is associated with positive treatment outcomes; however, there is an obvious need for systematic and high quality research designs to expand the evidence base for the behavioral treatment of MTVD.

In a randomized, blinded clinical trial, Pedrosa and colleagues (2016) evaluated the effectiveness of the Comprehensive Voice Rehabilitation Program (CVRP) compared with Vocal Function Exercises (VFEs) to treat FD. A total of 80 voice professionals presented with voice complaints for more than 6 months with a FD diagnosis were included in this study. Subjects were randomized into 2 voice treatment groups:

I. CVRP and
II. VFE.

I. CVRP and
II. VFE.

The rehabilitation program consisted of 6 voice treatment sessions and 3 assessment sessions performed before, immediately after, and 1 month after treatment. The outcome measures were self-assessment protocols (Voice-Related Quality of Life [V-RQOL] and Voice Handicap Index [VHI]), perceptual evaluation of vocal quality, and a visual examination of the larynx, both blinded. The randomization process produced comparable groups in terms of age, gender, signs, and symptoms. Both groups had positive outcome measures. The CVRP effect size was 1.09 for the V-RQOL, 1.17 for the VHI, 0.79 for vocal perceptual evaluation, and 1.01 for larynx visual examination. The VFE effect size was 0.86 for the V-RQOL, 0.62 for the VHI, 0.48 for the vocal perceptual evaluation, and 0.51 for larynx visual examination. Only 10 % of the patients were lost over the study. The authors concluded that both treatment programs were effective; and the probability of a patient improving because of the CVRP treatment was similar to that of the VFE treatment.

The study by Pedrosa et al (2016) was not informative because both groups received active treatment. Another study employing 2 active treatment groups compared in-person versus telephonic voice therapy reported a non-significant trend in improvement in both groups and no significant difference between the groups (Rangarathnam et al, 2015). A retrospective case-series study by Craig et al (2015) found no significant difference between voice therapy with and without adjunctive physical therapy for MTD. A short-term, small sample RCT (Watts et al, 2015a) found significantly greater improvement in vocal handicap, maximum phonation time, and acoustic measures of vocal function after participants received "stretch-and-flow voice therapy" compared to participants receiving vocal hygiene education alone. The authors concluded that "additional research incorporating larger samples will be needed to confirm and further investigate these findings". Furthermore, a small prospective uncontrolled series by the same investigator group (Watts et al, 2015b) also found improvements with stretch-and-flow voice therapy.

Resonant Voice Therapy

Barrichelo-Lindström and Behlau (2009) examined perceptually and acoustically Lessac's Y-Buzz and sustained productions of Brazilian Portuguese habitual /i/ vowels pre- and post-training and verified the presence of formant tuning and its association with the perception of a more resonant voice. The subjects of this study were 54 acting students (23 males and 31 females) with no voice problems, distributed in 7

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

groups. Each group received 4 weekly sessions of training. For the pre-training recording, they were asked to sustain the vowel /i/ in a habitual mode 3 times at self-selected comfortable frequencies and intensity. After training, they repeated the habitual /i/ and also the trained Y-Buzz.; 5 voice specialists rated how resonant each sample sounded. The fundamental frequency (F(0)), the first 4 formant frequencies, the distance between the frequencies of F(1) and F(0) were measured, as well as the harmonic frequency (H(2)) frequency and the difference between F(1) and H(2) in the case of male voices (Praat 4.4.33, Institute of Phonetic Sciences, University of Amsterdam, The Netherlands). The trained Y-Buzz was considered more resonant than the habitual /i/ samples, regardless the gender and demonstrated a lowering of the 4 formant frequencies. F(1) was especially lower in both groups (288 Hz for females and 285 Hz for males), statistically significant in the female group. The F(1)-F(0) difference was significantly smaller for the female Y-Buzz (52Hz), as well as F(1)-H(2) in the case of the male Y-Buzz (12Hz), suggesting formant tuning. The authors concluded that it was not possible to establish association between the perceptual grades and measures F(1)-F(0) or F(1)-H(2).

Hazlett and associates (2011) reviewed the current published available research into the impact of voice training on the vocal quality of professional voice users, and provided implications for vocal health and recommendations for further research. These investigators performed a systematic search of the literature using electronic databases and the following defined search terms: occupational voice or occupational dysphonia or voice and occupational safety and health. To obtain the comprehensive relevant literature, no studies were excluded on the basis of study design. A total of 10 studies that examined the impact of a voice training intervention on the vocal quality of professional voice users as a potential prevention strategy for voice disorders were selected for this review. The 10 studies ranged in design from observational to RCTs with mainly small sample sizes ($n = 11$ to 60); 9 studies showed that voice training significantly ($p < 0.05$) improved at least 1 voice-related measurement from the several investigated from baseline. A total of 5 studies reported that voice training significantly ($p < 0.05$) improved at least 1 measurement compared with no training. The authors concluded that the findings of this analysis indicated that there is no conclusive evidence that voice training improves the vocal effectiveness of professional voice users, as a result of a range of methodological limitations of the included studies. However, some studies showed that voice training significantly improved the knowledge, awareness, and quality of voice. Thus, there is a need for robust research to empirically confirm this, with implications for vocal health, and occupational safety and health policies.

Yiu and colleagues (2017) reviewed the literature on resonant voice therapy and evaluated the level of evidence on the effectiveness of using resonant voice therapy in treating dysphonia. Refereed journal papers from 1974 to 2014 were retrieved and reviewed by 2 independent reviewers using the keywords "humming, resonance, resonant voice, semi-occluded or closed tube phonation" using available database systems. Quality of evidence was evaluated by using the Grading of Recommendations Assessment, Development and Evaluation (GRADE). A total of 13 papers met the search criteria; 9 were selected by the 2 reviewers; 2 of the papers were RCTs and the other 7 were observational studies. At least 4 types of resonant voice therapies were described. They included the Lessac-Madsen resonant voice therapy, Y-Buzz, resonance therapy and humming. The overall level of quality of evidence was graded as "moderate". The authors concluded that there were limited studies that examined the effectiveness of resonant voice therapy. Most studies were small-scale uncontrolled observational studies with the inclusion of only small samples or specific populations. They stated that there is clearly a need for more large-scale RCTs with a wider range of populations to provide further evidence on the effectiveness of resonant voice training for different populations.

Essential Voice Tremor

Hyperkinetic dysarthria is characterized by abnormal involuntary movements affecting respiratory, phonatory, and articulatory structures impacting speech and deglutition.

Barkmeier- Kraemer and Clark (2018) reviewed the standard clinical evaluation and treatment approaches by SLPs for addressing impaired speech and deglutition in specific hyperkinetic dysarthria populations. A literature review was conducted using the data sources of PubMed, Cochrane Library, and Google Scholar. Search terms included

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

I. hyperkinetic dysarthria, essential voice tremor, voice tremor, vocal tremor, spasmodic dysphonia, spastic dysphonia, oromandibular dystonia, Meige syndrome, orofacial, cervical dystonia, dystonia, dyskinesia, chorea, Huntington's Disease, myoclonus; and evaluation/treatment terms:
II. Speech-Language Pathology, Speech Pathology, Evaluation, Assessment, Dysphagia, Swallowing, Treatment, Management, and diagnosis.

I. hyperkinetic dysarthria, essential voice tremor, voice tremor, vocal tremor, spasmodic dysphonia, spastic dysphonia, oromandibular dystonia, Meige syndrome, orofacial, cervical dystonia, dystonia, dyskinesia, chorea, Huntington's Disease, myoclonus; and evaluation/treatment terms:
II. Speech-Language Pathology, Speech Pathology, Evaluation, Assessment, Dysphagia, Swallowing, Treatment, Management, and diagnosis.

The standard SLP clinical speech and swallowing evaluation of chorea/Huntington's disease, myoclonus, focal and segmental dystonia, and essential vocal tremor typically includes

I. case history;
II. examination of the tone, symmetry, and sensorimotor function of the speech structures during non-speech, speech and swallowing relevant activities (i.e., cranial nerve assessment);
III. evaluation of speech characteristics; and
IV. patient self-report of the impact of their disorder on activities of daily living.

I. case history;
II. examination of the tone, symmetry, and sensorimotor function of the speech structures during non-speech, speech and swallowing relevant activities (i.e., cranial nerve assessment);
III. evaluation of speech characteristics; and
IV. patient self-report of the impact of their disorder on activities of daily living.

SLP management of individuals with hyperkinetic dysarthria includes behavioral and compensatory strategies for addressing compromised speech and intelligibility. Swallowing disorders are managed based on individual symptoms and the underlying pathophysiology determined during evaluation.

Novafon Local Vibration Voice Therapy for Dysphonia

Local vibration therapy is a non-invasive solution for muscle tensions, chronic pain and effects of a stroke. Results can be achieved especially in the fields of physiotherapy, occupational therapy, speech and language therapy and naturopathy. The most frequent application areas are the treatment of chronic muscle, joint and tendon diseases (e.g., arthritis, heel spur, myalgia, and tennis elbow). In the neuro-rehabilitation sector, spasticity, swallowing and voice disorders can be treated.

Latoszek (2020) examined the effectiveness of a 5-week Novafon local vibration voice therapy (NLVVT) program for the treatment of patients with dysphonia ($n = 11$). Treatment effects were assessed during (i.e., weekly) and after NLVVT. Large and significant treatment effects were revealed in acoustics (i.e., spectrography), and multi-parametric indices (i.e., Acoustic Voice Quality Index (AVQI), and DSI) during and after NLVVT (all p values < 0.01). Additionally, self-evaluation (i.e., VHI) showed a significant improvement after NLVVT ($p < 0.01$). Gender independent voice range profile parameters (i.e., acoustics) only showed significant effects after treatment ($p \leq 0.01$), but not during the treatment. Finally, aerodynamic measurement (i.e., phonation quotient) showed low treatment effects after NLVVT, which were non-significant ($p > 0.05$). The authors concluded that these preliminary findings showed that NLVVT might be successful in voice treatment. Large treatment effects might be expected in AVQI, DSI, spectrography and VHI after using NLVVT. Other voice characteristics showed smaller treatment effects (i.e., voice range profile parameters) or no meaningful treatment effects (i.e., phonation quotient). These preliminary results need to be further validated.

Dysphonic Caused by Benign Vocal Fold Lesions

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Schindler and colleagues (2012) noted that benign vocal fold lesions (BVFLs) are common in the general population, and have important public health implications and impact on patient QOL. Nowadays, phono-microsurgery is the most common treatment of these lesions. Voice therapy is generally associated in order to minimize detrimental vocal behaviors that increase the stress at the mid-membranous vocal folds. Nonetheless, the most appropriate standard of care for treating BVFL has not been established. These investigators analyzed voice changes in a group of dysphonic patients affected by BVFLs, evaluated with a multi-dimensional protocol before and after voice therapy. A total of16 consecutive patients, 12 women and 4 men, with a mean age of 49.7 years were enrolled. Each subject had 10 voice therapy sessions with an experienced speech/language pathologist for a period of 1 to 2 months, and was evaluated before and at the end of voice therapy with a multi-dimensional protocol that included self-assessment measures and video-stroboscopic, perceptual, aerodynamic and acoustic ratings. Video-stroboscopic examination did not reveal resolution of the initial pathology in any case. No improvement was observed in aerodynamic and perceptual ratings. A clear and significant improvement was visible on Wilcoxon signed-rank test for the mean values of Jitt %, Noise to Harmonic Ratio (NHR) and VHI scores. Even if it was possible that, for BVFLs, only a minor improvement of voice quality can be achieved after voice therapy, rehabilitation treatment still appeared useful as demonstrated by improvement in self-assessment measures. If voice therapy was provided as an initial treatment to the patients with BVFLs, this may lead to an improvement in the perceived voice quality, making surgical intervention unnecessary. The authors concluded that this was one of the first reports on the efficacy of voice therapy in the management of BVFLs; further studies are needed to confirm these preliminary findings. These researchers stated that this was a consecutive case-series study, and a control group (e.g., receiving only vocal hygiene programs) was not included. For this reason, the evidence level for this study was quite low (Level C according to the United Kingdom National Health Service) and it was not possible to demonstrate that the improvement in self-assessment measures was a consequence of rehabilitation treatment.

Zhuge and co-workers (2016) analyzed the voice characteristics of patients with early vocal fold polyps and examined the effects of voice therapy on patients. Voice therapy was conducted on 88 patients with early vocal fold polyps in an experimental group. Laryngostroboscopy, the VHI, and the DSI were applied to evaluate the patients' voice quality before and after treatment; 31 healthy volunteers also underwent evaluation of voice quality as a control group. The inter-group and intra-group differences in the results of laryngostroboscopy, VHI, and DSI were compared statistically. In the experimental group, 22 patients (25 %) withdrew from the treatment for various reasons. After voice therapy, the cure rate was 30.3 % (20/66). All the VHI values in the experimental group before treatment were statistically worse than the values in the control group. The P and T values of the VHI in the experimental group were still worse after treatment than the values in the control group, although the VHI did improve after treatment in the experimental group. The maximum phonation time (MPT), jitter, I-low, and DSI were statistically different between the control group and the experimental group both before and after treatment. Furthermore, the MPT, jitter, F0-high, I-low, and DSI improved after treatment in the experimental group. The authors concluded that patients with early vocal fold polyps had some degree of subjective and objective dysphonia; voice therapy could improve the voice quality in these patients. This was a case-control study with a relatively high drop-out rate (25 %) and a relatively low cure rate (30.3 %). Moreover, the cure rate may also have been skewed by the high drop-out rate.

Tibbetts and associates (2018) noted that vocal fold cysts are benign mid-membranous lesions of the true vocal fold, classified as mucus retention or epidermal inclusion cysts. Treatment is surgical excision with or without post-operative voice therapy. These investigators carried out a retrospective review of the demographics, treatment approach, and outcomes of patients treated for vocal fold cysts between 2009 and 2014; VHI-10 scores before and after treatment were compared using the Wilcoxon Rank-Sum test and the 2-tailed Student's t test. Video-stroboscopic examinations were reviewed for post-treatment changes in vibratory characteristics of the vocal folds. A total of 25 patients were identified, and 1 was excluded for incomplete records. Mean age was 41.9 years (66.7 % women), and mean follow-up time was 5.58 months. Micro-flap excision was pursued by 21/24 (87.5 %) patients, with 14 patients (58.3 %) undergoing peri-operative voice therapy; 1 cyst recurred; 2 patients elected for observation, and their cysts persisted. Overall, VHI-10 decreased from 23.8 to 6.6 ($p < 0.001$). There was a statistically significant reduction in VHI-10 in patients undergoing surgery with and without post-operative voice therapy ($p < 0.004$ and 0.001); however there was no significant difference between these 2 groups. Mucosal wave was classified as normal or

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

improved in the majority. Cysts were characterized as mucus retention cysts in 19/21 (90 %) and as epidermal inclusion cysts in 2/21 (10 %). The authors concluded that vocal fold cysts impacted mucosal wave and glottic closure. Surgical excision resulted in low rates of recurrence, and in improvement in the mucosal wave and VHI-10. Moreover, these researchers stated that peri-operative voice therapy did not offer a significant benefit. Mucus retention cysts were the majority, in contrast to other published studies.

Ogawa and Inohara (2018) updated their knowledge regarding the effectiveness of voice therapy for the treatment of vocal disturbance associated with BVFLs, including vocal cysts, nodules and polyps, and for the determination of the utility of voice therapy in treating organic voice disorders, while highlighting problems for the future development of this clinical field. These researchers carried out a review of the most recent literature on the therapeutic effects of voice therapy, vocal hygiene education or direct vocal training on vocal quality, the lesion appearance and discomfort felt by patients due to the clinical entity of BVFLs. Although voice therapy is principally indicated for the treatment of functional dysphonia without any organic abnormalities in the vocal folds, a number of clinicians have attempted to perform voice therapy even in dysphonic patients with BVFLs. The 2 major possible reasons for the effectiveness of voice therapy on vocal disturbance associated with BVFLs are hypothesized to be the regression of lesions and the correction of excessive/inappropriate muscle contraction of the phonatory organs. According to the current literature, a substantial proportion of vocal polyps certainly tend to shrink after voice therapy, but whether or not the regression results from voice therapy, vocal hygiene education or a natural cure is unclear at present due to the lack of controlled studies comparing 2 groups with and without interventions. Regarding vocal nodules, no studies have examined the effectiveness of voice therapy using proper experimental methodology. Vocal cysts are difficult to cure by voice therapy without surgical excision according to previous studies. The authors concluded that evidences at present is insufficient to support the use of voice therapy for the treatment of BVFLs.

White (2019) stated that BVFLs cause dysphonia by preventing full vocal fold closure, interrupting vibratory characteristics and increasing compensatory muscle tension. Management includes phono-surgery, voice therapy, pharmacotherapy or more commonly a combination of these interventions. This investigator presented current perspectives on the management of BVFLs, particularly exploring the role of voice therapy. This review highlighted variation in the management of BVFLs. There is evidence that phono-surgery is a well-tolerated and effective intervention for BVFLs. Primary voice therapy can frequently prevent surgery in vocal fold nodules and some types of polyps. Used as an adjunct to phono-surgery, pre-operative and post-operative voice therapy can improve patient-reported outcomes and acoustic parameters of the voice. However, heterogeneity of studies and poor descriptions of intervention components prevented a robust analysis of the impact of voice therapy. The author concluded that the current evidence consists of low-level studies using mixed etiology groups, which compromised internal and external validity. There are a few exceptions to this. Poor reporting and heterogeneous methodologies led to difficulties determining the components of a voice therapy intervention for this population. Consequently, the author was unable to evaluate, which intervention elements are beneficial to patients.

Pre-Operative Voice Therapy for Benign Vocal Fold Lesions

White et al (2023) noted that benign vocal fold lesions cause dysphonia by preventing vocal fold closure, causing irregular vibration and increasing compensatory muscle tension. Voice therapy delivered in addition to phono-surgery may improve voice and quality of life (QOL) outcomes; however, the evidence base is lacking and what constitutes voice therapy for this population is not defined. In a systematic review, these researchers examined the evidence for pre- and post-operative voice therapy to inform the development of an evidence-based intervention. Electronic databases were searched using key terms including dysphonia, phono-surgery, voice therapy and outcomes. Eligible articles were extracted and reviewed by the authors for risk of bias and for information regarding the content, timing and intensity of any pre- and post-operative voice therapy intervention. Of the 432 articles identified, 35 met the inclusion criteria and were included in the review -- 5 were RCTs, 2 were individual cohort studies, 1 was a case-control study and 26 were case-series studies. There was considerable heterogeneity in participant characteristics. Information was frequently lacking regarding the content timing and intensity of the reported voice therapy intervention, and where present, interventions were highly variable. The authors concluded that reporting in relevant literature

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

is limited in all aspects of content, timing and intensity of intervention. These researchers stated that further intervention development work is needed to develop a robust voice therapy treatment intervention for this population, before effectiveness work can commence.

In a qualitative interview study, White and Carding (2022) described factors influencing the content, timing, and intensity of pre- and post-operative voice therapy for patients undergoing phono-surgery for benign vocal fold lesions. These investigators also attempted to understand experts' rationale for decisions made; and analyzed factors influencing intervention in relation to the wider literature in order to contribute to the development of a complex intervention. Participants included 10 expert voice therapists with a mean of 22-year experience. Participants were asked to describe factors influencing their current practice and views on optimum treatment for patients undergoing phono-surgery for benign vocal fold lesions. Data were analyzed using the Framework Method of thematic analysis. Factors influencing intervention related to 4 key themes -- pathophysiological, patient, therapist, and service factors influenced the content, timing, and duration of the voice therapy provided. Consensus on core elements included delivering indirect and direct therapy pre-operatively to manage underlying causative factors and address patient expectations. Post-operative intervention focused on indirect therapy to facilitate wound healing and direct therapy to improve vibratory characteristics of the vocal fold. Elements of therapy were highly individualized within participants according to the 4 themes above; however, similarity between participants on broad parameters of intervention was high. The authors concluded that expert voice therapists use direct and indirect methods pre- and post-operatively to treat patients with benign vocal fold lesions. Optimizing wound healing and mobilization of the epithelium post-operatively are concerns for expert voice therapists which distinguish post-operative patients from other dysphonic patients. This study provided an insight into the factors influencing clinician's intervention provision that could contribute to the development of an optimal pre- and post-operative voice therapy intervention. In particular, these investigators stated that there is some literature suggesting the value of voice therapy to reduce the edema associated with vocal fold pathology especially in cases where lesions arise from vocal misuse. However, there is no strong evidence to support or refute this opinion which may explain the variation in pre-operative intervention. These researchers stated that there is a growing body of evidence that suggested that pre-operative voice therapy can negate surgical intervention in some cases. This now requires more robust scientific investigation to determine which aspects of voice therapy have the greatest potential to influence outcomes.

Voice Therapy for the Treatment of Sulcus Vocalis

Sulcus vocalis, defined as a type of groove along the free edge of the vocal fold, disrupts the normal, pliable vocal fold cover, causing alterations in the intrinsic mucosal wave. The primary symptom is breathy, effortful dysphonia. Schweinfurth, et al. (2020) explained that the primary goal of speech therapy in sulcus vocalis is to improve vocal efficiency. The method most commonly employed is direct speech therapy. When voice therapy is combined with external measures (e.g., amplification) and behavioral alterations (e.g., scheduling vocal rest periods), vocal fatigue may dissipate.

Rajasudhakar (2016) noted that sulcus vocalis is a structural deformity of the vocal ligament. It is the focal invagination of the epithelium deeply attaching to the vocal ligament. There is a dearth of literature on the outcome of voice therapy in sulcus vocalis condition. The primary objective of this study was to document voice characteristics of sulcus vocalis and the secondary objective was to establish the efficacy of voice therapy in a patient with sulcus vocalis. A trial of voice therapy was administered to the client who was diagnosed as having sulcus vocalis. Boon's facilitation techniques were used in voice therapy along with other techniques such as breath holding and push and pull approach prior to surgery. Acoustic, aerodynamic, perceptual, quantitative measures of voice quality and self-rating measurements were carried out before and after voice therapy. Improvement was noticed in 10/10 acoustic, 4/4 aerodynamic, perceptual, dysphonia severity index and voice handicap index scores, which hinted that voice therapy can be an option critically for clients with sulcus vocalis in the initial stage. The authors concluded that voice therapy showed promising improvement in the study and it must be recommended as the initial therapeutic option before any surgical management. These researchers stated that this study documented the immediate effect of voice therapy on multi-parametric approach in voice assessment. The long-term effect of the same can be studied

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

in future; they stated that more studies are needed in voice therapy outcome measures with more clients with sulcus vocalis.

Nam et al (2019) stated that lip trills are widely used as a voice warm-up technique among singers; however, little is known regarding the effects of lip trills in cases of voice disorders. These researchers examined the therapeutic effects of lip trills in patients with glottal gap. Patients with glottal gap were classified into 3 groups according to the type of gap: gap-only, gap with muscle tension dysphonia (MTD), and a sulcus vocalis group. Patients underwent perceptual, acoustic/aerodynamic analyses, stroboscopic evaluations, and subjective analyses using a questionnaire before and after lip trills. The results were analyzed before and after trills and according to and between the groups. The results in 42 patients were analyzed. Most of the parameters were improved and glottal gap was significantly reduced after trills in all patients. In the gap-only group (n = 19), most of the parameters showed improvement and were within the respective normal ranges, and glottal gap was improved after trills. In the MTD group (n = 13), although many parameters were improved, the improvement was not as prominent as in the gap-only group. In the sulcus vocalis group (n = 10), only some of the parameters were improved and the improvement in glottal gap was limited. The authors concluded that lip trills were an effective treatment for glottal gap. The therapeutic effect was prominent in the gap-only group, followed by the MTD and sulcus vocalis groups; and lip trills can be used as an adjuvant therapeutic option in voice therapy in cases of various voice disorders.

Voice Therapy for the Treatment of Vocal Fold Motion Impairment Following Chemotherapy Administration

Talmor et al (2021) stated that chemotherapy-induced vocal fold motion impairment (CIVFMI) is a rare complication of cancer therapy with potential for airway compromise. These investigators described 2 new cases of CIVFMI to add to the literature and characterized the demographics, symptoms, otolaryngologic examination findings, airway complication rates as well as prognosis of CIVFMI. They carried out searches of PubMed/Medline (1970 to May 1, 2020), Embase (1970 to May 1, 2020), and Cochrane Library using medical study heading (MeSH) terms related to chemotherapy (drug therapy, chemotherapy, vincristine, vinblastine, paclitaxel) and vocal cord motion impairment (vocal cord, cords, vocal folds, immobility, hypomobility). Exploratory pooling of data without formal meta-analysis was conducted. The preliminary search yielded 148 abstracts, review articles and studies. A total of 23 studies met inclusion criteria. There were 35 total cases presented in the literature, with a mean age of 29.5 (0.4 to 78). The most common cancer diagnosis was acute lymphoblastic leukemia (ALL; n = 15, 42.9 %), and the most common agent was vincristine (n = 30, 85.7 %). Dysphagia, bilateral CIVFMI, and vocal fold immobility rather than hypomobility were more common in pediatric patients. There were 8 cases of surgical airway intervention, including tracheostomy and posterior cordotomy. The duration of symptoms was 7 to 420 days, and spontaneous resolution was reported in 32 cases. The authors concluded that CIVFMI has potential for airway complications requiring surgical intervention. Spontaneous resolution following cessation of the offending agent is the most likely outcome. Bilateral CIVFMI, dysphagia and vocal fold immobility are more common in the pediatric population. Moreover, these researchers noted that the case presented in this study highlighted that symptoms of CIVFMI may present after cessation of chemotherapy. Early identification of CIVFMI and withdrawal of the offending agent remains essential in managing this disorder. They stated that further research is needed to examine the pathophysiology behind CIVFMI as descriptions regarding the mechanism of chemotherapy-induced neurotoxicity remain incomplete.

The authors stated that this study had several main drawbacks. First, CIVFMI is a rare complication, and the overall number of reported cases was low (n = 35). As such, it was difficult to draw convincing conclusions from trends noted in the data, such as the increased incidence of stridor and airway intervention in pediatric patients. Second, reporting of data was inconsistent. The latency period for CIVFMI was reported with a high degree of variability, making it difficult to report an accurate mean time from initiation of therapy to symptomatic onset. Third, patient symptoms were not completely reported in all cases. Two of the cases of unresolved CIVMFI had limited follow-up periods, making it difficult to interpret the significance of these findings. Fourth, cases of unilateral CIVFMI with mild symptoms may not have resulted in specialized evaluation of laryngeal function; thus, may have been under-reported in the literature.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Children with Vocal Fold Nodules

Adriaansen et al (2022) stated that vocal fold nodules (VFNs) are the main cause of pediatric dysphonia; voice therapy is recommended as the preferable therapeutic option for VFNs in children. In a systematic review, these investigators examined the available evidence on the effects of voice therapy in children with VFNs. This systematic literature review was developed following the PRISMA guidelines. The Cochrane Central Register of Controlled Trials (CENTRAL), Medline (via PubMed) and Embase were searched; and the grey literature was checked. The search strategy was based on 3concepts: VFNs, voice therapy and children. Two independent examiners determined study eligibility and extracted all relevant data from the included studies. The methodological quality of the included study was assessed using the QualSyst tool. By identifying, evaluating and summarizing the results of all relevant studies regarding voice therapy in pediatric VFNs, this systematic review made the available evidence more accessible to voice therapists, otolaryngologists and other relevant stakeholders. The authors concluded that a total of 24 studies were included in this systematic review – 8 studies (8/24) reported a significant improvement for at least 1 outcome parameter following voice therapy; however, 5 studies (5/24) did not report significant changes following voice therapy. All studies that did not test for significance (11/24) found improvements for 1 or more outcome parameters. The overall quality of the included studies was adequate (55 %). In summary, these researchers stated that there is some evidence that voice therapy is effective in children with VFNs; however, further well-designed studies, especially RCTs, are needed to confirm these findings.

Al-Kadi et al (2022) noted that one of the most prevalent pediatric ailments around the world is voice disorders. Approximately 5 million children suffer from voice disorders, and 3 out of 5 of them suffer from VFN-induced persistent dysphonia; and 19 out of 20 otolaryngologists recommend voice therapies for the treatment of pediatric VFNs. However, the benefits of these therapies still remain to be examined systematically. In a systematic review, these investigators examined the impact of voice therapy (direct and indirect) on pediatric patients with VFNs. In this systematic review of RCTs, 4 electronic databases, PubMed, CENTRAL (Cochrane), Science Direct, and Lancet, were examined for the literature survey. The impact of direct and indirect voice therapies on pediatric cases with VFNs was reviewed based on the results of the selected articles. Based on stringent inclusion and exclusion criteria, a total of 6 studies were selected. All these studies examined the effects of direct and indirect voice therapies on 2 types of voice disorders (i.e., dysphonia and vocal nodules). Only 1 of the 6 studies reported significant alleviation of the patient condition post-intervention. However, none of the studies discussed the clinical significance of the interventions. In addition, 3 of the 6 included studies used both direct and indirect voice therapies and reported substantial differences in the data collected before and after the interventions. However, overall, the studies reported more significant improvements in patient conditions. These researchers stated that more studies in this domain are still needed, especially to aid in understanding and defining the meaning of the term "effectiveness" with respect to voice therapies.

The authors stated that this trial had several drawbacks. First, only studies published in the English language were included and even the gray literature on this topic was ignored. Second, the number of studies included in this review was very small, which eliminated any odds of conducting a pooled meta-analysis to further validate the efficiencies of voice therapies. Third, the included studies were heterogeneous with respect to the research design and details surrounding the voice therapy employed. Fourth, the included studies lacked any standardization of the outcome measures, which made it difficult to draw any useful comparisons among the studies.

Transgender Women

In a systematic review and meta-analysis, Schwarz et al (2023) examined the available evidence on the effects of speech therapy and phonosurgery, for transgender women, in relation to the fundamental frequency gain of the voice, regarding the type of vocal sample collected, and compared the effectiveness of the treatments. Furthermore, the study design, year, country, types of techniques used, total therapy time, and vocal assessment protocols were analyzed. These investigators searched the PubMed, Lilacs, and SciELO databases for observational studies and clinical trials, published in English, Portuguese, or Spanish, between January 2010 and January 2023. The selection of studies was performed according to PRISMA 2020

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

guidelines. The quality of selected studies was assessed using the Newcastle-Ottawa scale. Of 493 studies, 31 were deemed potentially eligible and retrieved for full-text review, and 16 were included in the systematic review and meta-analysis. A total of 6 studies carried out speech therapy and 10 studies phonosurgery. The speech therapy time did not influence the post-treatment gain in voice fundamental frequency ($p = 0.6254$). The type of sample collected significantly influenced the post-treatment voice frequency gain ($p < 0.01$). When the vocal sample was collected via vowel ($p < 0.01$) and reading ($p < 0.01$), the gain was significantly more heterogeneous between the different types of treatment. Phonosurgery was significantly more effective in terms of fundamental frequency gain compared to speech therapy alone, regardless of the type of sample collected ($p < 0.01$). The average gain of fundamental frequency after speech therapy, in the /a/ vowel sample, was 27 Hz, 39.05 Hz in reading, and 25.42 Hz in spontaneous speech. In phonosurgery, there was a gain of 71.68 Hz for the vowel /a/, 41.07 Hz in reading, and 39.09 Hz in spontaneous speech. The study with the highest gain (110 Hz) collected vowels, and the study with the lowest gain (15 Hz), spontaneous speech. The majority of the included studies received a score between 4 and 8 on the Newcastle-Ottawa Scale. The authors concluded that the type of vocal sample collected influenced the gain result of the fundamental frequency after treatment. Speech therapy and phonosurgery increased the fundamental frequency and improved female voice perception and vocal satisfaction; however, phonosurgery yielded a greater fundamental frequency gain in the different samples collected. Moreover, these researchers stated that the quality of evidence of the studies was low, given the lack of RCTs, the small sample sizes, and the different methods of collecting fundamental frequency; therefore, this study may provide additional evidence on the role of voice feminization therapy and phonosurgery on voice feminization and encourage further research in the field. These investigators stated that the drawbacks of this systematic review were the small number of published articles, their small sample sizes, as well as the differences in the fundamental frequency analysis methods of each study. As these researchers verified differences between the different types of collected voice samples, it was not possible to group all the results without considering these differences.

Tracheoesophageal Voice Therapy in Post-Laryngectomy Rehabilitation

An UpToDate review of alaryngeal speech production (Lewin, et al., 2024) stated that there are three major approaches used to restore oral communication after total laryngectomy: the artificial larynx (electrolarynx), tracheoesophageal voice restoration, and esophageal speech. The authors explained that esophageal speech production depends on air that is introduced into the esophagus and is then expelled past the pharyngoesophageal segment, the vibratory source for sound production. The authors stated that the primary disadvantages of developing esophageal speech are the length of time required to learn the technique and the reduction in voice quality compared with tracheoesophageal and normal laryngeal speech, compounded by the lack of availability of trained clinicians who can teach it.

Sparks et al (2023) noted that following total laryngectomy, surgical voice restoration is considered the optimal modality for re-establishing communication via trachea-esophageal voice. Yet beyond the insertion of a voice prosthesis to elicit voice production, there is sub-optimal clinical knowledge of how to rehabilitate the perceptual quality of trachea-esophageal voice. In a systematic review, these investigators examined the quality and effectiveness of therapeutic interventions for trachea-esophageal voice. These researchers stated that the findings of this review will inform the development of a novel trachea-esophageal voice therapy intervention. They performed a systematic literature review in accordance with the PRISMA guidelines. A total of 8 electronic databases were searched using a pre-specified search strategy. Records were independently screened by 2 reviewers against inclusion and exclusion criteria. Eligible studies were assessed for quality using the PEDro, ROBIN-T, and NHLBI critical appraisal tools. Data were extracted pertaining to participant characteristics as well as the content, dosage, intensity, and outcomes of interventions. A total of 6,344 records were identified, of which 38 were included for full-text review; and 6 studies met the eligibility criteria for inclusion. Voice rehabilitation was not the primary focus in the majority of studies, and the risk of bias was identified across studies. There was significant heterogeneity in the interventions and outcome measures used within studies with insufficient detail provided on intervention content for trachea-esophageal voice. Evidence for the effectiveness of interventions was limited and inconsistent across studies. The authors concluded that this review found that trachea-esophageal voice therapy is an under-researched area of clinical practice. Evidence from the small body of existing studies was not sufficiently robust to inform clinical practice at this time. These researchers stated that this review

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

highlighted the need to develop and test interventions aimed at improving the perceptual quality of trachea-esophageal voice. Moreover, these investigators noted that further studies are needed to examine multi-dimensional aspects of alaryngeal voice production such as pitch, resonance, and functional voice use; alongside wider investigation of the effectiveness of breathing and articulation approaches observed in the included studies. No studies considered the views of service users in detail within the development, study design, or intervention content and this is identified of an area of research importance.

The authors stated that this review aimed to identify evidence for therapeutic rehabilitation of trachea-esophageal voice and to critically appraise the quality of the evidence and effectiveness of interventions. The review showed that there is a very small evidence base in this area, identifying only 6 studies, which presented risk of bias. Due to the heterogeneity of interventions and outcomes used it was not possible to conduct a meta-analysis. The review found limited information on what therapeutic interventions may be appropriate for trachea-esophageal voice. Moreover, the small number of studies and identified risk of bias further reduce confidence in the findings and generalizability within clinical practice.

References

The above policy is based on the following references:

1. Adriaansen A, Meerschman I, Van Lierde K, D'haeseleer E. Effects of voice therapy in children with vocal fold nodules: A systematic review. Int J Lang Commun Disord. 2022;57(6):1160-1193.
2. Ahern R, Lippincott LH, Wisdom G. Voice rehabilitation after laryngectomy: An overview. J La State Med Soc. 2002;154(3):118-120.
3. Al-Kadi M, Alfawaz MA, Alotaibi FZ. Impact of voice therapy on pediatric patients with dysphonia and vocal nodules: A systematic review. Cureus. 2022;14(4):e24433.
4. Alegria R, Freitas SV, Manso MC. Effectiveness of voice therapy in patients with vocal fold nodules: A systematic search and narrative review. Eur Arch Otorhinolaryngol. 2020;277(11):2951-2966.
5. Altman KW, Atkinson C, Lazarus C. Current and emerging concepts in muscle tension dysphonia: A 30-month review. J Voice. 2005;19(2):261-267.
6. Andrews S, Warner J, Stewart R. EMG biofeedback and relaxation in the treatment of hyperfunctional dysphonia. Br J Disord Commun. 1986;21(3):353-369.
7. Angsuwarangsee T, Morrison M. Extrinsic laryngeal muscular tension in patients with voice disorders. J Voice. 2002;16(3):333-343.
8. Barcelos CB, Silveira PAL, Guedes RLV, et al. Multidimensional effects of voice therapy in patients affected by unilateral vocal fold paralysis due to cancer. Braz J Otorhinolaryngol. 2018;84(5):620-629.
9. Barkmeier-Kraemer J, Lato A, Wiley K. Development of a speech treatment program for a client with essential vocal tremor. Semin Speech Lang. 2011;32(1):43-57.
10. Barkmeier-Kraemer JM, Clark HM. Speech-language pathology evaluation and management of hyperkinetic disorders affecting speech and swallowing function. Tremor Other Hyperkinet Mov (N Y). 2017;7:489.
11. Barrichelo-Lindström V, Behlau M. Resonant voice in acting students: Perceptual and acoustic correlates of the trained Y-Buzz by Lessac. J Voice. 2009;23(5):603-609.
12. Birkent H, Akçam T, Gerek M, Ertaş I, Ozkaptan Y. [Results of voice therapy in functional voice disorders]. Kulak Burun Bogaz Ihtis Derg. 2004;12(5-6):120-127.
13. Bos-Clark M, Carding P. Effectiveness of voice therapy in functional dysphonia: Where are we now? Curr Opin Otolaryngol Head Neck Surg. 2011;19(3):160-164.
14. Branski RC, Murray T. Voice therapy. eMedicine Otolaryngology and Facial Plastic Surgery Topic 683. Omaha, NE: eMedicine.com; updated August 4, 2005.
15. Buddiga P. Vocal cord dysfunction. eMedicine Allergy and Immunology. New York, NY: eMedicine; updated August 4, 2010.
16. Carding P. Evidence based voice therapy. Autumn 2011 Research Seminars in the Speech-Language Pathology Unit, Karolinska Institute, Stockholm, Sweden, November 29, 2011.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

17. Carding PN, Horsley IA, Docherty GJ. A study of the effectiveness of voice therapy in the treatment of 45 patients with nonorganic dysphonia. J Voice. 1999;13(1):72-104.
18. Casper JK, Murry T. Voice therapy methods in dysphonia. Otolaryngol Clin North Am. 2000;33(5):983-1002.
19. Chiara T, Martin D, Sapienza C. Expiratory muscle strength training: Speech production outcomes in patients with multiple sclerosis. Neurorehabil Neural Repair. 2007;21(3):239-249.
20. Craig J, Tomlinson C, Stevens K, et al. Combining voice therapy and physical therapy: A novel approach to treating muscle tension dysphonia. J Commun Disord. 2015;58:169-178.
21. da Cunha Pereira G, de Oliveira Lemos I, Dalbosco Gadenz C, Cassol M. Effects of voice therapy on muscle tension dysphonia: A systematic literature review. J Voice. 2018;32(5):546-552.
22. Dejonckere PH. Clinical implementation of a multidimensional basic protocol for assessing functional results of voice therapy. A preliminary study. Rev Laryngol Otol Rhinol (Bord). 2000;121(5):311-313.
23. Desjardins M, Halstead L, Cooke M, Bonilha HS. A systematic review of voice therapy: What "effectiveness" really implies. J Voice. 2017;31(3):392.e13-392.e32.
24. Dromey C, Nissen SL, Roy N, Merrill RM. Articulatory changes following treatment of muscle tension dysphonia: Preliminary acoustic evidence. J Speech Lang Hear Res. 2008;51(1):196-208.
25. Eastwood C, Madill C, Mccabe P. The behavioural treatment of muscle tension voice disorders: A systematic review. Int J Speech Lang Pathol. 2015;17(3):287-303.
26. Fex B, Fex S, Shiromoto O, Hirano M. Acoustic analysis of functional dysphonia: Before and after voice therapy (accent method). J Voice. 1994;8(2):163-167.
27. Hazlett DE, Duffy OM, Moorhead SA. Review of the impact of voice training on the vocal quality of professional voice users: Implications for vocal health and recommendations for further research. J Voice. 2011;25(2):181-191.
28. Higgins MB, Chait DH, Schulte L. Phonatory air flow characteristics of adductor spasmodic dysphonia and muscle tension dysphonia. J Speech Lang Hear Res. 1999;42(1):101-111.
29. Hocevar-Boltezar I, Janko M, Zargi M. Role of surface EMG in diagnostics and treatment of muscle tension dysphonia. Acta Otolaryngol. 1998;118(5):739-743.
30. Houtz DR, Roy N, Merrill RM, Smith ME. Differential diagnosis of muscle tension dysphonia and adductor spasmodic dysphonia using spectral moments of the long-term average spectrum. Laryngoscope. 2010;120(4):749-757.
31. Kao YC, Chen SH, Wang YT, et al. Efficacy of voice therapy for patients with early unilateral adductor vocal fold paralysis. J Voice. 2017;31(5):567-575.
32. Kim H-T. Vocal feminization for transgender women: Current strategies and patient perspectives. Int J Gen Med. 2020;13:43-52.
33. Kitzing P, Akerlund L. Long-time average spectrograms of dysphonic voices before and after therapy. Folia Phoniatr (Basel). 1993;45(2):53-61.
34. Kooijman PG, de Jong FI, Oudes MJ, et al. Muscular tension and body posture in relation to voice handicap and voice quality in teachers with persistent voice complaints. Folia Phoniatr Logop. 2005;57(3):134-147.
35. Latoszek BBV. Preliminary study of Novafon local vibration voice therapy for dysphonia treatment. Logoped Phoniatr Vocol. 2020;45(1):1-9.
36. LeBorgne WD, Donahue EN. Voice therapy as primary treatment of vocal fold pathology. Otolaryngol Clin North Am. 2019;52(4):649-656.
37. Lee EK, Son YI. Muscle tension dysphonia in children: Voice characteristics and outcome of voice therapy. Int J Pediatr Otorhinolaryngol. 2005;69(7):911-917.
38. Lewin JS, Hoffman HT, Sperry SM. Alaryngeal speech rehabilitation. UpToDate [online serial]. Waltham, MA: UpToDate; updated June 2024.
39. Lowell SY, Kelley RT, Colton RH, et al. Position of the hyoid and larynx in people with muscle tension dysphonia. Laryngoscope. 2012;122(2):370-377.
40. Maryn Y, De Bodt MS, Van Cauwenberge P. Ventricular dysphonia: Clinical aspects and therapeutic options. Laryngoscope. 2003;113(5):859-866.
41. Mathieson L, Hirani SP, Epstein R, et al. Laryngeal manual therapy: A preliminary study to examine its treatment effects in the management of muscle tension dysphonia. J Voice. 2009;23(3):353-366.
42. Mathieson L. The evidence for laryngeal manual therapies in the treatment of muscle tension dysphonia. Curr Opin Otolaryngol Head Neck Surg. 2011;19(3):171-176.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

43. Merck & Co., Inc. Vocal cord nodules (singer's nodules). In: Merck Manual of Diagnosis and Therapy. 17th ed. MH Beers, R Berkow, eds. White House Station, NJ: Merck; 1997.
44. Miller S. Voice therapy for vocal fold paralysis. Otolaryngol Clin North Am. 2004;37(1):105-119.
45. Morrison MD, Nichol H, Rammage LA. Diagnostic criteria in functional dysphonia. Laryngoscope. 1986;96(1):1-8.
46. Morrison MD, Rammage LA, Belisle GM, et al. Muscular tension dysphonia. J Otolaryngol. 1983;12(5):302-306.
47. Morrison MD, Rammage LA. Muscle misuse voice disorders: Description and classification. Acta Otolaryngol. 1993;113(3):428-434.
48. Nam I-C, Kim S-Y, Joo Y-H, et al. Effects of voice therapy using the lip trill technique in patients with glottal gap. J Voice. 2019;33(6):949.e11-949.e19.
49. National Institutes of Health, National Institute on Deafness and Other Communication Disorders (NIDCD). Vocal cord paralysis. NIDCD Health Information. Bethesda, MD: NIDCD; June 1999. Available at: http://www.nidcd.nih.gov/health/voice/vocalparal.asp. Accessed July 26, 2002.
50. National Jewish Medical and Research Center. Vocal cord dysfunction. Medfacts. Denver, CO: National Jewish; updated July 23, 2002. Available at: http://www.nationaljewish.org/medfacts/vocal.html. Accessed July 26, 2002.
51. Nolan IT, Morrison SD, Arowojolu O, et al. The role of voice therapy and phonosurgery in transgender vocal feminization. J Craniofac Surg. 2019;30(5):1368-1375.
52. Nguyen DD, Kenny DT, Tran ND, Livesey JR. Muscle tension dysphonia in Vietnamese female teachers. J Voice. 2009;23(2):195-208.
53. Ogawa M, Inohara H. Is voice therapy effective for the treatment of dysphonic patients with benign vocal fold lesions? Auris Nasus Larynx. 2018;45(4):661-666.
54. Patel RR, Liu L, Galatsanos N, Bless DM. Differential vibratory characteristics of adductor spasmodic dysphonia and muscle tension dysphonia on high-speed digital imaging. Ann Otol Rhinol Laryngol. 2011;120(1):21-32.
55. Pedersen M, McGlashan J. Surgical versus non-surgical interventions for vocal cord nodules. Cochrane Database Syst Rev. 2001;(2):CD001934.
56. Pedrosa V, Pontes A, Pontes P, et al. The effectiveness of the comprehensive voice rehabilitation program compared with the vocal function exercises method in behavioral dysphonia: A randomized clinical trial. J Voice. 2016;30(3):377.e11-e19..
57. Rajasudhakar R. Effect of voice therapy in sulcus vocalis: A single case study. S Afr J Commun Disord. 2016;63(1):e1-e5.
58. Rangarathnam B, McCullough GH, Pickett H, et al. Telepractice versus in-person delivery of voice therapy for primary muscle tension dysphonia. Am J Speech Lang Pathol. 2015;24(3):386-399.
59. Rosen CA, Murry T, Zinn A, et al. Voice handicap index change following treatment of voice disorders. J Voice. 2000;14(4):619-623.
60. Roy N, Bless DM, Heisey D, Ford CN. Manual circumlaryngeal therapy for functional dysphonia: An evaluation of short- and long-term treatment outcomes. J Voice. 1997;11(3):321-331.
61. Roy N, Ferguson NA. Vowel formant changes following manual circumlaryngeal therapy for functional dysphonia: Evidence of laryngeal lowering? J Med Speech-Lang Pathol. 2001;9(3):169-175.
62. Roy N, Ford CN, Bless DM. Muscle tension dysphonia and spasmodic dysphonia: The role of manual laryngeal tension reduction in diagnosis and management. Ann Otol Rhinol Laryngol. 1996;105(11):851-856.
63. Roy N, Hendarto H. Revisiting the pitch controversy: Changes in speaking fundamental frequency (SFF) after management of functional dysphonia. J Voice. 2005;19(4):582-591.
64. Roy N, Leeper HA. Effects of the manual laryngeal musculoskeletal tension reduction technique as a treatment for functional voice disorders: Perceptual and acoustic measures. J Voice. 1993;7(3):242-249.
65. Roy N, Merrill RM, Gray SD, Smith EM. Voice disorders in the general population: Prevalence, risk factors, and occupational impact. Laryngoscope. 2005;115(11):1988-1995.
66. Roy N, Nissen SL, Dromey C, Sapir S. Articulatory changes in muscle tension dysphonia: Evidence of vowel space expansion following manual circumlaryngeal therapy. J Commun Disord. 2009;42(2):124-135.
67. Roy N, Whitchurch M, Merrill RM, et al. Differential diagnosis of adductor spasmodic dysphonia and muscle tension dysphonia using phonatory break analysis. Laryngoscope. 2008;118(12):2245-2253.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

68. Roy N. Differential diagnosis of muscle tension dysphonia and spasmodic dysphonia. Curr Opin Otolaryngol Head Neck Surg. 2010;18(3):165-170.
69. Roy N. Functional dysphonia. Curr Opin Otolaryngol Head Neck Surg. 2003;11(3):144-148.
70. Rubin JS, Blake E, Mathieson L. Musculoskeletal patterns in patients with voice disorders. J Voice. 2007;21(4):477-484.
71. Rubin JS, Lieberman J, Harris TM. Laryngeal manipulation. Otolaryngol Clin North Am. 2000;33(5):1017-1034.
72. Ruotsalainen JH, Sellman J, Lehto L, et al. Interventions for treating functional dysphonia in adults. Cochrane Database Syst Rev. 2007;(3):CD006373.
73. Sama A, Carding PN, Price S, et al. The clinical features of functional dysphonia. Laryngoscope. 2001;111(3):458-463.
74. Samlan RA, Webster KT. Swallowing and speech therapy after definitive treatment for laryngeal cancer. Otolaryngol Clin North Am. 2002;35(5):1115-1133.
75. Sapienza CM, Walton S, Murry T. Adductor spasmodic dysphonia and muscular tension dysphonia: Acoustic analysis of sustained phonation and reading. J Voice. 2000;14(4):502-520.
76. Schindler A, Mozzanica F, Ginocchio D, et al. Vocal improvement after voice therapy in the treatment of benign vocal fold lesions. Acta Otorhinolaryngol Ital. 2012;32(5):304-308.
77. Schwarz K, Cielo CA, Spritzer PM, et al. A speech therapy for transgender women: An updated systematic review and meta-analysis. Syst Rev. 2023;12(1):128.
78. Schweinfurth J. Sulcus vocalis. Otolaryngology and Facial Plastic Surgery. Medscape Drugs & Diseases. New York, NY: Medscape; updated July 17, 2020.
79. Sparks F, Coffey M, Dipper L, et al. Tracheoesophageal voice therapy in postlaryngectomy rehabilitation: A systematic review. J Voice. 2023 Nov 23 [Online ahead of print].
80. Speyer R. Effects of voice therapy: A systematic review. J Voice. 2008;22(5):565-580.
81. Stachler RJ, Francis DO, Schwartz SR, et al. Clinical practice guideline: Hoarseness (dsphonia) (Update). Otolaryngol Head Neck Surg. 2018;158(1_suppl):S1-S42.
82. Stepp CE, Merchant GR, Heaton JT, Hillman RE. Effects of voice therapy on relative fundamental frequency during voicing offset and onset in patients with vocal hyperfunction. J Speech Lang Hear Res. 2011;54(5):1260-1266.
83. Talmor G, Nguyen B, Geller MT, et al. Vocal fold motion impairment following chemotherapy administration: Case reports and review of the literature. Ann Otol Rhinol Laryngol. 2021;130(4):405-415.
84. Tibbetts KM, Dominguez LM, Simpson CB. Impact of perioperative voice therapy on outcomes in the surgical management of vocal fold cysts. J Voice. 2018;32(3):347-351.
85. Trajano FMP, Almeida LNA, de Alencar SAL, et al. Group voice therapy reduces anxiety in patients with dysphonia. J Voice. 22020;34(5):702-708.
86. van Gogh CD, Verdonck-de Leeuw IM, Boon-Kamma BA, et al. The efficacy of voice therapy in patients after treatment for early glottic carcinoma. Cancer. 2006;106(1):95-105.
87. Van Houtte E, Claeys S, D'haeseleer E, et al. An examination of surface EMG for the assessment of muscle tension dysphonia. J Voice. 2013;27(2):177-186.
88. Van Houtte E, Van Lierde K, Claeys S. Pathophysiology and treatment of muscle tension dysphonia: A review of the current knowledge. J Voice. 2011;25(2):202-207.
89. Van Lierde KM, De Bodt M, Dhaeseleer E, et al. The treatment of muscle tension dysphonia: A comparison of two treatment techniques by means of an objective multiparameter approach. J Voice. 2010;24(3):294-301.
90. Van Lierde KM, De Ley S, Clement G, et al. Outcome of laryngeal manual therapy in four Dutch adults with persistent moderate-to-severe vocal hyperfunction: A pilot study. J Voice. 2004;18(4):467-474.
91. Watts CR, Diviney SS, Hamilton A, et al. The effect of stretch-and-flow voice therapy on measures of vocal function and handicap. J Voice. 2015b;29(2):191-199.
92. Watts CR, Hamilton A, Toles L, et al. A randomized controlled trial of stretch-and-flow voice therapy for muscle tension dysphonia. Laryngoscope. 2015a;125(6):1420-1425.
93. White A. Management of benign vocal fold lesions: Current perspectives on the role for voice therapy. Curr Opin Otolaryngol Head Neck Surg. 2019;27(3):185-190.
94. White AC, Awad R, Carding P, et al. Pre and post-operative voice therapy intervention for benign vocal fold lesions: A systematic review. J Voice. 2023;37(6):857-874.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

95. White AC, Carding P. Pre- and postoperative voice therapy for benign vocal fold lesions: Factors influencing a complex intervention. J Voice. 2022;36(1):59-67.
96. Yiu EM, Lo MC, Barrett EA. A systematic review of resonant voice therapy. Int J Speech Lang Pathol. 2017;19(1):17-29.
97. Yorkston KM, Hakel M, Beukelman DR, Fager S. Evidence for effectiveness of treatment of loudness, rate, or prosody in dysarhria: A systematic review. Centre for Reviews and Dissemintation (CRD). Database of Abstracts of Reviews of Effects (DARE). Accession No. 12007009331. York, UK: University of York; December 1, 2008.
98. Young N, Blitzer A. Management of supraglottic squeeze in adductor spasmodic dysphonia: A new technique. Laryngoscope. 2007;117(11):2082-2084.
99. Zeitels SM, Casiano RR, Gardner GM, et al. Management of common voice problems: Committee report. Otolaryngol Head Neck Surg. 2002;126(4):333-348.
100. Zheng YQ, Zhang BR, Su WY, et al. Laryngeal aerodynamic analysis in assisting with the diagnosis of muscle tension dysphonia. J Voice. 2012;26(2):177-181.
101. Zhuge P, You H, Wang H, et al. An analysis of the effects of voice therapy on patients with early vocal fold polyps. J Voice. 2016;30(6):698-704.

Policy History

- Last Review opens in a new browser pop-up window 12/08/2025

  Effective: 09/17/2002

  Next Review: 07/09/2026

- Review History opens in a new browser pop-up window

- Definitions opens in a new browser pop-up window

Additional Information

close popup

- Clinical Policy Bulletin Notes opens in a new browser pop-up window

You are now leaving the Aetna website.

Links to various non-Aetna sites are provided for your convenience only. Aetna Inc. and its subsidiary companies are not responsible or liable for the content, accuracy, or privacy practices of linked sites, or for products or services described on these sites.

Continue opens in new window

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF